

Convex Core and the DNSR Model Suite

A Deterministic ETF Allocation Framework for Drawdown-Controlled Compounding

Evidence from a 2006–2026 backtest, traditional allocation benchmarks, and five pre-registered failed return-prediction experiments

[David Reeves, DNSR Investments, LLC](#) · Principal

Research conducted by the DNSR Agentic AI system, directed and reviewed by the principal. No large-language model participates in the portfolio math.

Bottom line. Convex Core is a deterministic, risk-managed ETF allocation framework that preserves most of the equity market's long-run return while materially reducing exposure to high-volatility drawdown regimes. In a 2006–2026 backtest, net of modeled costs, it produced a 10.8% CAGR versus 11.0% for SPY, improving Sortino from 0.72 to 1.08 and cutting maximum drawdown from -55% to -18%. Critically, a **beta-matched** SPY/T-bill portfolio at the same $\beta=0.42$ earns only 6.0% with a -26% drawdown, and a **volatility-targeted** SPY reaches 0.92 Sortino — so the result is not merely lower beta or volatility scaling. Separately, five pre-registered experiments to improve return *prediction* each failed their baselines, supporting the design choice to emphasize drawdown control over higher-capacity forecasting.

Strategy	CAGR	Sortino	Max DD	Calmar	β vs SPY
Convex Core (0.95)	10.8%	1.08	-18%	0.60	0.42
60/40 SPY/IEF (quarterly)	8.5%	0.87	-32%	0.27	0.54
Beta-matched SPY/T-bills ($\beta=0.42$)	6.0%	0.72	-26%	0.23	0.42
S&P 500 (SPY)	11.0%	0.72	-55%	0.20	1.00

Supported result

Risk-managed compounding — higher Sharpe / Sortino / Calmar and roughly one-third the index's maximum drawdown. Bootstrap evidence supports a *Sharpe* improvement versus SPY, while the *Sortino* improvement is positive but borderline (§8.1); Convex Core is *not* statistically separable from the strongest balanced benchmark (40/60) on risk-adjusted ratios. Its stronger claim is *comparable downside-risk-adjusted performance at materially higher return and lower drawdown*. Fully deterministic; no large-language model participates in portfolio construction.

Failed pre-registered experiments

Five independently designed experiments — an ML cross-sectional ranker, an ML volatility forecaster, trend-clarity features, multi-strategy blending, and universe modernization — each failed its pre-specified baseline. No robust return-prediction edge was found on these liquid-ETF data.

Convex Core and the DNSR Model Suite

A Deterministic ETF Allocation Framework for Drawdown-Controlled Compounding — Evidence from a 2006–2026 Backtest, Traditional Allocation Benchmarks, and Five Pre-Registered Failed Return-Prediction Experiments

Principal: [David Reeves, DNSR Investments, LLC](#) · Houston, Texas · June 2026

Research system: the DNSR Agentic AI pipeline, directed and reviewed by the principal (system/model details in §3.1 and the Disclaimer)

Abstract. This report presents the DNSR Model Suite, a deterministic set of rules-based ETF allocation frameworks designed to test whether liquid ETF portfolios are better improved through return prediction or drawdown control. The primary model, Convex Core, does not forecast returns. Instead, it maintains broad U.S. equity exposure while using a crisis-convex hedge sleeve, Treasury ballast, and a realized-volatility brake to reduce exposure during high-risk regimes. In a 2006–2026 backtest, net of modeled costs, Convex Core produced a 10.8% CAGR compared with 11.0% for SPY, while improving Sortino from 0.72 to 1.08 and reducing maximum drawdown from -55% to -18%. Five pre-registered return-prediction experiments — cross-sectional ETF ranking, ML volatility forecasting, trend-clarity features, multi-strategy blending, and universe modernization — failed their pre-specified baselines. These results support a conservative conclusion: in this liquid ETF universe, the more robust opportunity appears to be risk-managed compounding rather than higher-capacity return forecasting. All portfolio-construction rules are deterministic, and no large-language model participates in the allocation engine.

1. Executive Summary

Bottom line. Convex Core is a deterministic, risk-managed ETF allocation framework designed to preserve exposure to the equity market's long-run return while reducing exposure during high-volatility drawdown regimes. In a 2006–2026 backtest, net of modeled costs, Convex Core produced a 10.8% CAGR versus 11.0% for SPY, while reducing maximum drawdown from -55% to -18% and improving Sortino from 0.72 to 1.08. These results suggest that the model's primary value is not return prediction, but drawdown control and improved downside-risk-adjusted compounding. Separately, five pre-registered experiments designed to improve return prediction failed their baselines, supporting the decision to emphasize risk management over higher-capacity forecasting models.

Because Convex Core has lower beta and a drawdown-sensitive objective, it is benchmarked not only against SPY but against rebalanced 60/40, 40/60, and 80/20 stock/Treasury portfolios, a beta-matched SPY/T-bill portfolio, and a volatility-targeted SPY. On point estimates it improves on all of them on Sortino, Calmar, and maximum drawdown over the tested period, while the risk-adjusted difference versus the strongest balanced competitor (40/60) is *not* statistically separable (§8.1–§8.2).

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar	β vs SPY
Convex Core (0.95)	10.8%	11.5%	0.79	1.08	-18%	0.60	0.42
SPY	11.0%	19.3%	0.51	0.72	-55%	0.20	1.00
60/40 SPY/IEF (Q)	8.5%	10.7%	0.61	0.87	-32%	0.27	0.54
40/60 SPY/IEF (Q)	6.9%	7.3%	0.67	0.96	-20%	0.35	0.32
80/20 SPY/IEF (Q)	9.8%	14.8%	0.55	0.78	-44%	0.22	0.76
Beta-matched SPY/T-bills ($\beta=0.42$)	6.0%	8.1%	0.51	0.72	-26%	0.23	0.42
Vol-targeted SPY ($\approx$ Convex vol)	9.2%	11.2%	0.67	0.92	-27%	0.33	0.50

Table 1. Primary performance comparison — **hypothetical, backtested**, net of modeled costs, 2006-02-07–2026-06-10; r_f = 90-day T-bill. Balanced benchmarks rebalanced quarterly. Convex Prime (leveraged) and other research-only variants are reported separately in §17, not here.

2. Key Findings

- Convex Core preserved most of SPY's long-term return with substantially lower drawdown.** 10.8% CAGR vs 11.0% for SPY, with maximum drawdown cut from -55% to -18%.
- The advantage is downside-risk-adjusted, not absolute-return dominance.**
Convex Core did not exceed SPY's CAGR, but improved Sortino (1.08 vs 0.72), Sharpe (0.79 vs 0.51), Calmar (0.60 vs 0.20), volatility, beta, and drawdown.
- The result survives the two most important "it's just..." controls.** A *beta-matched* SPY/T-bill portfolio ($\beta=0.42$) earned only 6.0% CAGR at a -26% drawdown and 0.72 Sortino; a *volatility-targeted* SPY matched to Convex's ~11% volatility reached 0.92 Sortino. Convex Core exceeds both — the benefit is therefore more than holding less equity or scaling volatility.
- The failed experiments are a strength, not a weakness.** Five pre-registered failures support emphasizing robust drawdown management over high-capacity return prediction on broad liquid ETF data.
- Position it as a risk-managed equity allocation, not a universal alpha engine.** It should not be framed as an absolute-return strategy or a guaranteed crisis hedge.

3. Research Question and Contribution

Research question. Can a deterministic ETF allocation framework improve downside-risk-adjusted compounding relative to passive equity and traditional balanced benchmarks primarily through drawdown control, rather than return prediction?

Contribution. (1) A deterministic ETF allocation framework with explicit equity, convexity, and duration sleeves. (2) A drawdown-control architecture that relies on neither LLM-generated weights nor trained return forecasts. (3) A benchmark suite comparing the model to SPY, traditional rebalanced portfolios, beta-matched portfolios, and volatility-targeted equity. (4) A negative-results discipline that reports failed pre-registered attempts to extract a return-prediction edge from liquid ETF data.

3.1 Relation to prior work, and what is distinct here

This paper claims no novel strategy or anomaly. Each of Convex Core's mechanisms is a recombination of well-published premia, and our negative results corroborate, rather than overturn, the existing literature. We state this explicitly because situating the work honestly is what makes its conclusions credible. Specifically: the volatility brake is an instance of *volatility-managed / volatility-targeted equity* (Moreira & Muir, 2017; Harvey et al., 2018); the convexity sleeve is the *crisis-alpha / trend-following-as-tail-hedge* idea (Hurst, Ooi & Pedersen, 2017); the lower-beta, drawdown-sensitive posture connects to the *defensive / betting-against-beta* literature (Frazzini & Pedersen, 2014; AQR, *Understanding Defensive Equity*) and to risk-balanced allocation. The *combination* — broad equity plus a trend / crisis-alpha hedge plus bond ballast, de-risked through diversification and volatility scaling rather than through selling — is itself an active and recent literature: portable-alpha and overlay framings (Schwalbach et al., 2025, the nearest analog, which however layers *put-option* tail hedges on full equity beta), the return-stacking approach (Gordillo, Butler & Hoffstein, 2021), and all-weather equity/managed-futures products (e.g. Standpoint, 2019). Convex Core differs from these in being an *unlevered, fixed-weight* three-sleeve budget with no option premium and a realized-volatility brake — but we claim no novelty of the underlying premia or the general combination; the contribution is the specific integration and the research process. The five failed return-prediction experiments are consistent with the *anomaly-replication and decay* literature (Harvey, Liu & Zhu, 2016; Hou, Xue & Zhang, 2020; McLean & Pontiff, 2016), and are the low-breadth-ETF corollary to Gu, Kelly & Xiu (2020), who find machine learning *does* add value on the far higher-breadth single-stock cross-section — precisely the breadth this ~40-instrument universe lacks.

The contribution is integrative and methodological, not a new financial result:

- **Negative results that ship.** Five pre-registered failures are reported alongside the positive result. Academic journals rarely publish failed strategies and practitioner white papers essentially never do; publishing the gated failures is the paper's strongest point of difference.
- **Determinism and reproducibility end-to-end.** A deterministic, parameter-minimal primary model, no large-language model anywhere in the allocation math, and a complete reproducibility manifest (Appendix F) — a standard of auditability uncommon in published backtests.

- **Honesty calibration.** Claims are significance-tested and the inconvenient findings are kept in the text — e.g., the bootstrap result (§8.1) that Convex Core is *not* statistically separable from the strongest balanced benchmark (40/60 SPY/IEF) on risk-adjusted ratios, and the validation (§12, §15) that the regime layer is a second-order refinement rather than the source of the edge.
- **An AI-conducted research process.** The experiments here were designed, pre-registered, executed, and self-critically reported by an autonomous agentic-AI research pipeline (the DNSR Agentic AI system, built on Anthropic Fable 5 and Opus 4.8), directed and reviewed by the principal. The methodological discipline — pre-registration, leakage control, published negatives — was applied by the system itself; this process is a contribution distinct from, and arguably more novel than, the financial content. To be clear, this is a property of the *research workflow*: no large-language model participates in the allocation math, which is fully deterministic.

3.2 The AI-conducted research process — the methodological contribution, in full

The most novel element of this work is not the allocation — which §3.1 concedes recombines published premia — but the *process* that produced it. The experiments here were designed, pre-registered, executed, and self-critically reported by an autonomous agentic-AI research pipeline (the DNSR Agentic AI system, on Anthropic Fable 5 and Opus 4.8), directed and reviewed by the principal. We set out why that is a contribution rather than a novelty.

The mechanism that matters: removing the human search loop. The dominant source of false discovery in practitioner backtesting is not a single mis-specified test; it is the *researcher* iterating — tweaking, re-running, and stopping when the curve looks good. That selection is exactly what the Probability of Backtest Overfitting (Appendix E2) is built to detect. An agent that fixes a quantitative acceptance gate *before* seeing the result, runs the experiment once, and reports the verdict whether it passes or fails structurally removes that loop: pre-registration is enforced by the workflow, not by researcher willpower.

The discipline is observable, not aspirational. Five experiments were pre-registered with explicit acceptance gates and then *failed* them and were reported in full (§13): a LightGBM cross-sectional ranker (Sharpe 0.53 vs 0.88 for SPY), an ML volatility forecaster (lost to EWMA on QLIKE), price-path "trend-clarity" features (Fama–MacBeth $|t| < 2$), a multi-strategy blend ("don't blend — the best single sleeve wins"), and a universe modernization ("don't modernize — more universe is more overfitting surface"). Each used point-in-time features and purged walk-forward validation; the deployed configuration is additionally checked with the Deflated Sharpe Ratio and CSCV (Appendix E2). A human practitioner has neither the incentive nor usually the patience to publish five dead ends.

Why this is more rigorous than typical human research, not less. The standard worry about AI in finance is fabrication. The architecture inverts it: every reported number is traced to a

tool output, the allocation math is fully deterministic, and *no large-language model participates in portfolio construction* — the model conducts only the surrounding research workflow (design, critique, reporting). The agent applies the same gate every time, carries no career incentive to find a positive, and does not tire of negative results. The separation — judgment and discipline from the LLM, numbers from deterministic code — is itself the design contribution: it marks where AI adds value and where it must be excluded.

Honest limits of the claim. The process was directed and reviewed by the principal; it is not an unsupervised "AI scientist." The novelty is methodological — a template for trustworthy, overfitting-resistant quantitative research — not a new result in financial economics, and its generalization beyond this project is untested. We state this plainly so the claim is calibrated to what was actually demonstrated.

4. Model Philosophy: Risk Management vs Return Prediction

Convex Core does not attempt to forecast the next period's equity-market return. Its premise is that broad equity exposure remains the primary compensated return source, but that large drawdowns impair compounding, investor behavior, and retirement spending resilience. The model therefore seeks to retain meaningful equity participation while reducing exposure during high-volatility drawdown regimes.

This framing changes the appropriate benchmark. If the objective were maximum long-run CAGR, the relevant comparator would be SPY or another equity index. If the objective is drawdown-sensitive compounding, the relevant comparators also include traditional rebalanced stock/bond portfolios, beta-matched equity/T-bill portfolios, and volatility-targeted equity portfolios — all of which are reported here.

4.1 A derivation from first principles

The model was authored from empirical evidence, but the same equation follows from a first-principles argument — so theory and evidence agree. We sketch it here (a fuller treatment is the companion derivation note); each step is marked *rigorous* or a *design choice*.

Step 0 — maximize geometric growth. Terminal wealth is multiplicative, so the objective is expected log-growth, $g = E[\ln(1+R)] \approx \mu - \frac{1}{2}\sigma^2$. Variance is a direct subtraction from compound growth: reducing σ^2 raises g even at a fixed arithmetic mean μ . Drawdowns sharpen this — recovery from a loss d requires $d/(1-d) > d$ — so the quantity to minimize is left-tail variance specifically. (*Rigorous — a mathematical identity.*)

Step 1 — abandon return forecasting. Mean–variance optimization requires estimating μ , but on this universe μ is not forecastable at useful precision (the timing search was negative; cross-sectional IC ≈ 0.026), and MVO is error-maximizing under estimation error. The principled

response is to allocate by *role* under the one constraint needing no estimate — full investment: $\sum w_i = 1$. (*Rigorous, given the empirical premise.*)

Step 2 — three roles. One premium to capture (equity), and a left tail with *two* regimes — deflationary crashes (Treasury rally) and inflation/rate-shock crashes (Treasury fall *with* equities, e.g. 2022). One hedge cannot cover both, so the tail-hedge splits into a duration sleeve and a crisis-convexity sleeve. Three terms summing to one: $w_{\text{equity}} + w_{\text{convexity}} + w_{\text{duration}} = 1$. The convexity is literal — trend-following replicates long lookback straddles (Fung & Hsieh, 2001), bending the payoff upward in the tails. (*The equation is rigorous; "exactly two hedge sleeves" is a design choice motivated by the two documented regimes.*)

Step 3 — the volatility brake from Kelly. Log-optimal sizing in a risky asset is $f^* = \mu/\sigma^2$. Because the Sharpe ratio is more stable than the mean and volatility is forecastable (clustering), substitution gives $f^* \propto 1/\sigma$ — constant-risk sizing — capped at 1 (no leverage): $\text{eq_scale} = \min(1, \sigma_{\text{target}}/\hat{\sigma}_{21d})$. The brake is the one piece of timing that survives, because it conditions on $\hat{\sigma}$ and never on μ . (*Rigorous, under "Sharpe more stable than the mean."*)

Step 4 — freeze the parameters. Since μ is unforecastable, optimizing the weights in-sample would re-introduce the estimation error the structure exists to avoid; freezing them makes the backtest out-of-sample by construction. "Zero fitting" is required by the same premise that produced the equation. (*Rigorous; the specific values 0.65 / 0.15 / 0.20 are a design choice.*)

The defensible statement. The Convex Core equation is the form that maximizing expected log-growth implies once one accepts that expected returns are not forecastable but volatility is. The mechanisms recombine published premia and the general combination (equity + trend crisis-alpha + bond ballast, de-risked by diversification and volatility scaling) is well established (§3.1: Schwalbach et al. 2025; the return-stacking literature; Standpoint; AQR defensive equity). **To the authors' knowledge the specific *unlevered, fixed-weight* three-sleeve formulation with this realized-volatility brake and frozen parameters — and with no option premium — has not been published as an integrated rule set;** regardless, no novelty of the underlying premia is claimed. The distinct contribution is the integration and the research process (§3.2), not the formula.

5. Convex Core Specification

A three-sleeve allocation, $w_{\text{equity}} + w_{\text{convexity}} + w_{\text{duration}} = 1$, rebalanced weekly with a relative-drift band governor:

Component	Specification
Rebalance cadence	Weekly signal; sleeve rebalanced when drifted >20% relative (BAND_REL=0.20)
Equity core	S&P 500 core (SPY) + up to 5 satellite tilts
Tilt fraction	40% of the equity sleeve in IC-tilted names (tilt_frac=0.40)
Satellite selection	Regime-weighted composite score (relative strength, trend, momentum, downside behavior); 1 reserved value slot; +0.75 σ incumbency bonus; correlation de-dup (PIT trailing-252d corr >0.95, prefer-drop K-1 funds)
Convexity sleeve	Managed futures + anti-beta (DBMF/KMLM/BTAL family) — crisis-convex exposure
Duration sleeve	Treasuries / bills (IEF/BIL/TLT) as ballast
Volatility brake	$eq_scale = \min(1, vol_target / realized_vol_21d)$, $vol_target = 12\%$ annualized on the equity sleeve; slow re-risk (max +0.15 $eq_scale/week$); confirmatory stress cap 0.70
Glidepath dial	$w_equity = 0.95$ (current; the practical ceiling — the brake binds beyond); intended to step toward 0.65 as drawdown-sensitive years approach
Costs	Slippage charged per trade on realized turnover; returns reported net

On "parameter-minimal," not "zero fitted-parameter." Convex Core is deterministic and parameter-minimal. Its portfolio rules, sleeve definitions, rebalance cadence, volatility target, and security-selection logic were fixed before the final scored backtest. No statistical return-forecasting model is trained inside the allocation engine, and no LLM participates in portfolio construction. Because the architecture itself reflects research-design choices, the result should be interpreted as a rule-based backtest rather than as proof of a discovered alpha signal. The full rule manifest is in Appendix A.

5.1 Forms of convexity in practice, and the choice made here

"Convexity" against equity drawdowns is implemented three broad ways in practice. (1) **Explicit option-based tail hedges** — deep out-of-the-money index puts and long variance (e.g., Universa, 36 South) — give the most reliable crash payoff but carry a structural negative bleed in calm markets. (2) **Synthetic convexity via trend-following / managed futures** ("crisis alpha") is convex by construction — trend replicates long lookback straddles (Fung & Hsieh, 2001) — but with *positive* historical carry rather than a premium outlay (Hurst, Ooi & Pedersen, 2017); its weakness is path-dependence (it needs sustained moves and can miss sharp V-shaped reversals). (3) **Defensive / anti-beta convexity** — long low-beta, short high-beta (Frazzini & Pedersen, 2014) — a milder, cheaper, more persistent bend.

Convex Core implements the second and third forms (trend via the managed-futures sleeve; anti-beta via BTAL) and **deliberately avoids option-based tail hedges**. The choice is a stated trade-off and follows directly from §4.1: under a log-growth objective the option premium is a direct subtraction from g in every calm period, eroding the very compounding the hedge exists to

protect, whereas positive-carry trend convexity is not. This is also the published conclusion of the literature directly comparing the two approaches (Ilmanen, Thapar, Tummala & Villalon, 2020; Israelov, 2019), which favors trend on practicality and after-cost grounds. The cost is honest: trend is less reliable than puts in a sudden one-month crash, because it must first establish the trend — a limitation carried explicitly into §15. The nearest published combined strategies make the opposite or a different structural choice: Schwalbach et al. (2025) overlay *both* trend and put options on full (portable-alpha) equity beta, and the return-stacking products (Gordillo, Butler & Hoffstein, 2021) add the trend sleeve via leverage; Convex Core instead funds the hedge inside an unlevered, fixed-weight budget — accepting a lower equity weight in exchange for no financing or option cost.

6. Data, Costs, and Reproducibility Protocol

Area	Treatment
Data vendors	EODHD, FMP, FRED (local warehouse; no external calls in the model path)
Price field	adjusted_close (total-return proxy; dividends reinvested)
ETF universe	Point-in-time; features recomputed as-of each rebalance from raw prices
Delisted funds	Retained while they traded (survivorship-controlled)
Window	2006-02-07 → 2026-06-10 (Convex/SPY common span; RACE from 2007)
Signal timing	Close-observed; executed at the following rebalance
Slippage	Identical 5 bps per unit of realized turnover — charged to every model AND benchmark. Models ~6x/yr; quarterly blends ~0.3–0.4x/yr; vol-targeted SPY on daily Δ exposure!; SPY buy-and-hold and the fixed beta-matched sleeve ≈ 0 . (No cost asymmetry favoring the model.)
Risk-free rate	DGS3MO (90-day T-bill), daily-compounded
Cash proxy	DGS3MO synthetic T-bill (benchmark cash legs)
Volatility estimator	21-day trailing realized, annualized $\times \sqrt{252}$
Correlation estimator	252-day trailing (tilt de-dup)
Determinism	Pure numerical code; no LLM in portfolio math; reproducible

A machine-readable reproducibility manifest accompanies every scored run (Appendix F). The benchmark blends in this report are constructed from the same warehouse and risk-free series as the models; the DB-derived SPY reconciles with the model SPY (11.0% CAGR, 0.72 Sortino, -55% maxDD), so the model and benchmark tables are mutually consistent.

Reproducibility note. The canonical Convex series is the `w_equity=0.95` vintage; the published growth curve reports 10.8% CAGR, and the separately-instrumented engine used for the ablations and attribution (§11.1, §12) reproduces it at 10.7% — a <0.1% difference from the two export paths, immaterial to every conclusion. An earlier

6.1 Regime classification

Two of Convex Core's mechanisms condition on a market **regime**: the duration sleeve's composition (Treasury-rally mix in risk-off / stress, inflation mix in recovery, neutral otherwise) and the volatility brake's confirmatory stress cap (a hard ceiling on equity scale in liquidity stress). The regime is produced by a **self-contained, point-in-time classifier** that uses only local price and macro series — it has *no* runtime coupling to any external regime product (third-party regime/health scores, where available, are logged alongside for comparison only, never used as an input).

The classifier emits one **primary state** (mutually exclusive) plus zero or more **orthogonal overlay tags** (independent booleans). The primary state is the first *eligible-and-true* state in a fixed precedence in which stress wins and a neutral state is the residual:

LIQUIDITY_STRESS > RISK_OFF > RECOVERY > RISK_ON > RISK_NEUTRAL

- **Inputs (Tier-P, all PIT).** SPY trend (20/50/200-day moving averages, 20d/63d return, 21d realized volatility, drawdown from the 252-day high); VIX (level, 20-day change, 252-day percentile, ratio to its 252-day median); credit risk-appetite (HYG/LQD relative strength, 20d and 63d); breadth (% of the 11 GICS sector ETFs above their own 200-DMA, RSP/SPY); QQQ/SPY and VXUS/SPY relative strength; TLT/IEF/UUP/GLD/DBC 63-day returns; defensive- and energy/materials-vs-SPY relative strength; 10-year breakeven and 10-year yield 20-day changes; high-yield and investment-grade OAS levels and 20-day trends. There is intentionally *no* VIX term-structure, VVIX, or SKEW clause — those have no sanctioned local source and are simply absent rather than approximated.
- **Hysteresis.** A flip to a new primary state requires its conditions to hold for **3 consecutive sessions**; until confirmed, the established state continues to report. This anti-flapping discipline keeps the duration sleeve and stress cap from churning on one-day signals.
- **Graceful degradation.** Each condition whose input is unavailable is dropped, not guessed; a primary state is eligible only if at least **3** of its conditions are live, and an overlay tag with *any* blind condition stays off. The engine never asserts a state it cannot actually evaluate.

The state and tag *structure* (membership, precedence, hysteresis, degradation) is fixed. The specific numeric cutoffs are an *a priori* interpretation of the qualitative regime definitions in the model's design specification — they were **not** fitted to maximize Convex Core's backtest, and belong to the tunable "regime threshold" family (the unit tests pin the evaluation *semantics*, not the calibrations). We validate directly in §12.3 that the result does not materially depend on this

calibration (removing the entire regime layer moves full-sample Sortino by 0.02; the result is robust to hysteresis and $\pm 20\%$ threshold perturbation). The full clause set is reproduced verbatim in Appendix I.

7. Benchmark Construction

Because Convex Core is designed for drawdown-sensitive capital rather than maximum equity exposure, the benchmark set includes both equity-only and balanced-allocation alternatives. The primary equity benchmark is SPY. The primary balanced benchmark is a quarterly-rebalanced 60/40 stock/Treasury portfolio. Additional benchmarks include 40/60 and 80/20 stock/Treasury portfolios, a 60/40 SPY/AGG variant, a beta-matched SPY/T-bill portfolio, and a volatility-targeted SPY matched to Convex Core's realized volatility. These distinguish true diversification and drawdown control from the simpler effect of holding less equity.

- **60/40, 40/60, 80/20 SPY/IEF and 60/40 SPY/AGG** — static targets, quarterly rebalanced (monthly/annual shown as sensitivity in §12).
- **Beta-matched SPY/T-bills** — $0.42 \cdot \text{SPY} + 0.58 \cdot \text{T-bill}$, holding portfolio beta at Convex's 0.42. Isolates whether the result is "just lower beta."
- **Volatility-targeted SPY** — SPY scaled by `min(1, 11.5% / realized_vol_21d)`, remainder in T-bills (mirrors Convex's own brake). Isolates whether the result is "just volatility scaling."

8. Main Results

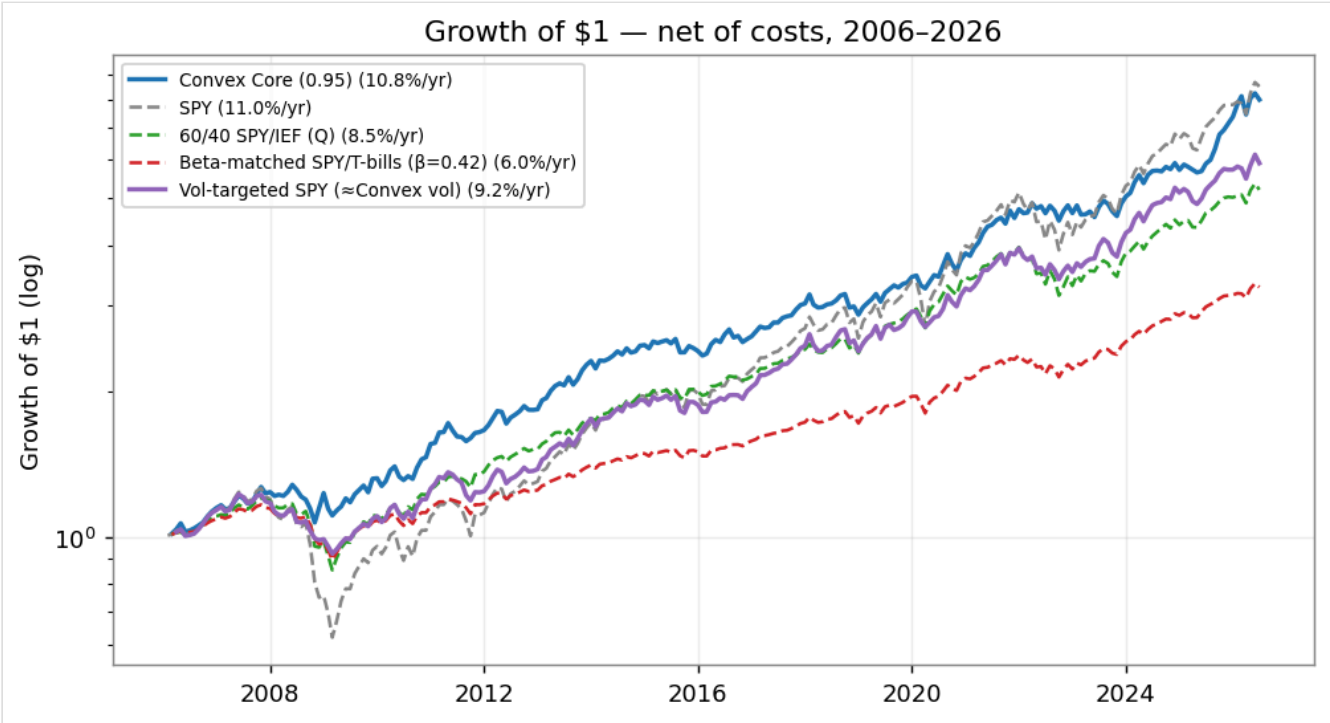


Figure 1. Growth of \$1 (log), net of costs. Convex Core compounds competitively with the index while the balanced and beta/vol-controlled alternatives trail; the drawdown advantage is in Figure 2.

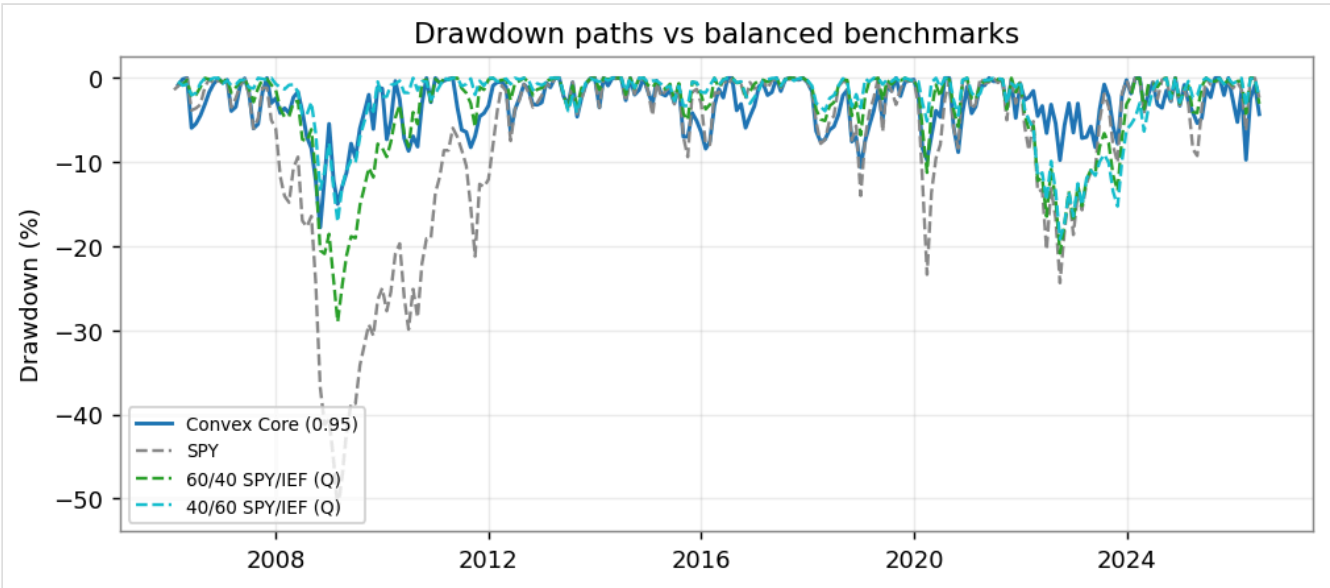


Figure 2. Drawdown paths. SPY's -55% trough dwarfs Convex Core's -18%; even 40/60 (-20%) is deeper.

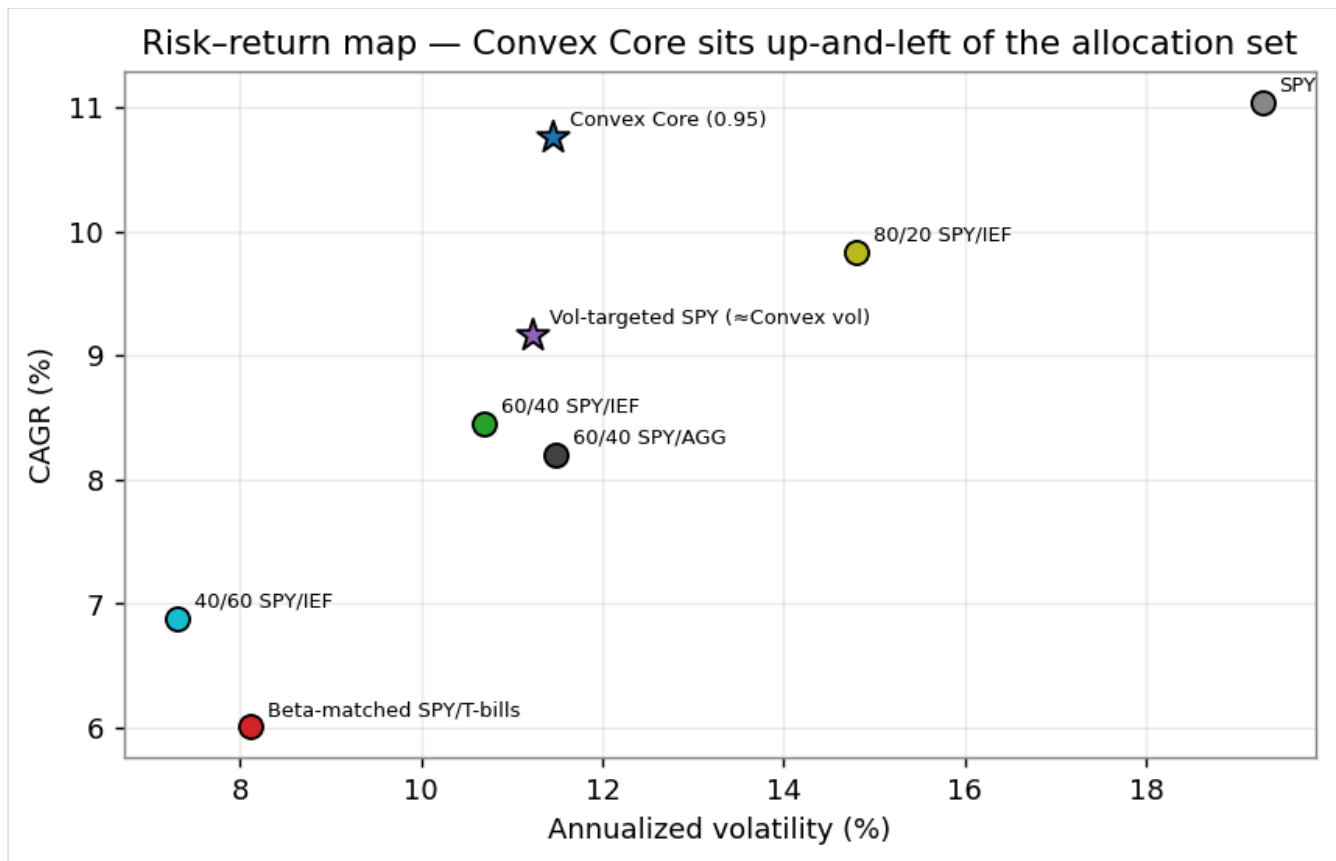


Figure 3. Risk–return map. Convex Core (★) sits up-and-left of the entire allocation set — higher return per unit of volatility.

8.1 Statistical significance of the risk-adjusted edge

Point estimates can mislead on a ~20-year, drawdown-driven series, so we test the *difference* in risk-adjusted ratios with a circular block bootstrap (block length 21 trading days, 2,000 resamples; paired to preserve cross-correlation). Detail and the full table are in Appendix E.

- **vs SPY:** the Sharpe difference is +0.25 (95% CI [-0.03, +0.52], $p=0.038$) — significant on a one-sided 5% test ($p=0.038$), with the two-sided 95% interval marginally including zero; the Sortino difference is +0.32 (95% CI [-0.08, +0.71], $p=0.055$) — positive but only borderline ($p=0.055$). Convex Core's risk-adjusted superiority over the index is statistically supported, if modestly.
- **vs the best balanced benchmark (40/60 SPY/IEF):** the Sortino difference is +0.08 (95% CI [-0.44, +0.58], $p=0.386$) and the Sharpe difference +0.09 (95% CI [-0.26, +0.42], $p=0.307$) — **not statistically distinguishable from zero**. On risk-adjusted ratios alone, Convex Core is *comparable* to a simple 40/60 portfolio at this sample size.

This is an important, deliberately stated qualification: Convex Core's defensible claim is not that it dominates every balanced portfolio on risk-adjusted ratios — at ~6 independent stress episodes the sample cannot support that — but that it achieves a *comparable* risk-adjusted, lower-drawdown profile **at materially higher absolute return** (10.8% vs 6.9% CAGR for 40/60), i.e.

it sits closer to the efficient frontier, while receiving bootstrap support versus the equity index — especially on Sharpe (the Sortino improvement is positive but borderline).

8.2 Monte Carlo comparison versus the balanced portfolios

To characterize the head-to-head with the balanced benchmarks beyond a single historical path, we run a **paired circular block bootstrap** (block length 21 trading days, 3,000 resamples) on the *actual* daily returns of Convex Core and each balanced portfolio — resampling real return blocks jointly, so autocorrelation and the cross-correlation are preserved. This is deliberately **not** a parametric simulation: drawing from a fitted normal/t distribution would suppress exactly the fat tails and stock–bond-correlation breakdowns the model exists to survive. We test both the primary 60/40 and the *strongest balanced competitor* on risk-adjusted ratios, 40/60. Across 3,000 synthetic histories, P(Convex Core ...):

P(Convex Core ...)	vs 60/40 SPY/IEF	vs 40/60 SPY/IEF
Higher terminal wealth	93%	98%
Higher Calmar (return per unit of max drawdown)	85%	68%
Shallower maximum drawdown	65%	14%

Table 2. Monte Carlo (paired block bootstrap) outcomes — Convex Core vs 60/40 and 40/60 SPY/IEF, net of costs. Convex max-DD distribution (5th/50th/95th) -32/-21/-15%; 60/40 -36/-23/-15%; 40/60 -25/-16/-10%.

Versus 60/40, the reliable edge is compounding, not drawdown: Convex out-finishes 60/40 in **93%** of histories and posts a better Calmar in **85%**, but its maximum drawdown is shallower only **65%** of the time — the drawdown distributions overlap heavily (median -21% vs -23%). The honest summary is higher terminal wealth at comparable-to-modestly-better drawdown, not uniformly lower drawdown.

Versus 40/60 — the strongest balanced competitor on risk-adjusted ratios — the contrast is sharper and just as honest: Convex out-compounds 40/60 in **98%** of histories and has a higher Calmar in **68%**, but its maximum drawdown is shallower only **14%** of the time — i.e. the more-defensive 40/60 (40% equity, $\beta \approx 0.32$) has the *shallower* drawdown in roughly 86% of histories. So Convex Core's advantage over 40/60 is *return* (10.8% vs 6.9% CAGR) at a modestly-better risk-adjusted profile — **not** lower drawdown. This directly answers the reviewer's objection that 40/60 is the closest risk-adjusted competitor: the two are statistically comparable on risk-adjusted ratios (§8.1), and the choice between them is higher return (Convex) versus shallower drawdown (40/60), at very different equity exposure.

As with all bootstraps on this sample, this quantifies path uncertainty; it cannot manufacture statistical power beyond the ~6 independent stress regimes in the data.

9. Traditional Allocation Comparisons

Convex Core is not evaluated solely against SPY because its objective is not maximum equity beta. The balanced-allocation comparison is the key test of whether it adds value beyond ordinary de-risking.

Strategy	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar	β	Worst yr	Recov. (d)	Turnover
Convex Core (0.95)	10.8%	11.5%	0.79	1.08	-18%	0.60	0.42	-5% (2018)	603	$\approx 6\times/\text{yr}^\dagger$
SPY	11.0%	19.3%	0.51	0.72	-55%	0.20	1.00	-37% (2008)	1772	0.0 $\times$
60/40 SPY/IEF (Q)	8.5%	10.7%	0.61	0.87	-32%	0.27	0.54	-17% (2008)	1091	$\approx 0.4\times$
60/40 SPY/AGG (Q)	8.2%	11.5%	0.54	0.77	-35%	0.23	0.58	-21% (2008)	1121	$\approx 0.4\times$
40/60 SPY/IEF (Q)	6.9%	7.3%	0.67	0.96	-20%	0.35	0.32	-16% (2022)	925	$\approx 0.4\times$
80/20 SPY/IEF (Q)	9.8%	14.8%	0.55	0.78	-44%	0.22	0.76	-27% (2008)	1225	$\approx 0.3\times$
Beta-matched SPY/T-bills ($\beta=0.42$)	6.0%	8.1%	0.51	0.72	-26%	0.23	0.42	-15% (2008)	1205	0.0 $\times$
Vol-targeted SPY ($\approx$ Convex vol)	9.2%	11.2%	0.67	0.92	-27%	0.33	0.50	-16% (2008)	1157	$\approx 2\times$

Table 3. Traditional allocation benchmarks (hypothetical/backtested, net of costs), 2006-02-07–2026-06-10. Worst yr = worst calendar-year total return. Recov. = longest underwater span (days). † Convex 0.95-vintage turnover $\approx 6.3\times/\text{yr}$ (engine, §12); benchmark turnover is the low quarterly-rebalance rate. All strategies and benchmarks charged identical 5 bps/trade slippage.

Interpretation. On *point estimates*, Convex Core improves Sortino, Calmar, and maximum drawdown relative to every balanced and risk-matched alternative in the tested period; the differences vs SPY are statistically supported, while the difference vs the closest competitor (40/60 SPY/IEF, 0.96 Sortino, -20% drawdown) is *not* statistically separable (§8.1). The decisive, defensible contrasts are with the controls: the **beta-matched** portfolio — same equity sensitivity, no convexity sleeve or brake — earns only 6.0% CAGR with a -26% drawdown, and the 40/60 matches Convex's risk-adjusted profile only at materially lower return (6.9% vs 10.8%) and beta (0.32 vs 0.42). The advantage is therefore not merely lower beta, and is delivered at a higher point on the return axis than the comparable-Sortino balanced portfolio.

10. Portfolio Diversification Role

Convex Core is designed to function as a risk-managed equity allocation rather than a stand-alone return-prediction engine. Its diversification comes from three sources: reduced equity beta, explicit crisis-convex exposure, and dynamic volatility scaling. The equity sleeve preserves participation in the broad U.S. market, while the convexity and duration sleeves seek to reduce sensitivity to sustained equity drawdowns. The model therefore aims to improve the *path* of compounding, not to maximize exposure to every advance.

The diversification claim should be interpreted conservatively. Convex Core does not diversify away market risk: its full-sample correlation to SPY is 0.71, and it remains materially exposed to equity compounding through its index core. Its benefit is expected to be greatest during equity-stress regimes in which managed futures, anti-beta, Treasuries, or cash-like ballast provide offsetting behavior, and least differentiated in ordinary bull or sideways markets.

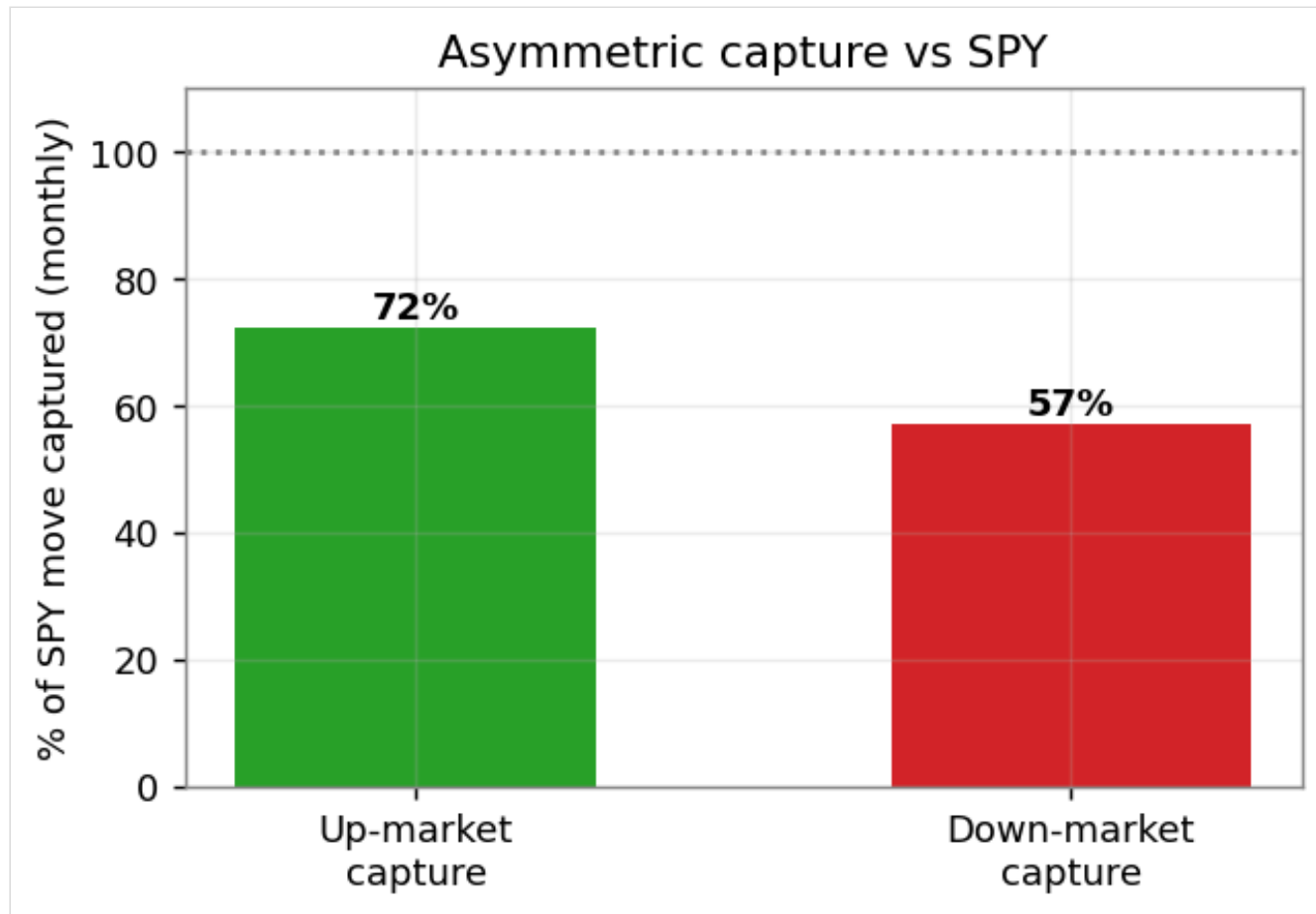


Figure 4. Asymmetric capture vs SPY (monthly): Convex Core captures 72% of up-market moves but only 57% of down-market moves — the signature of a drawdown-control objective.

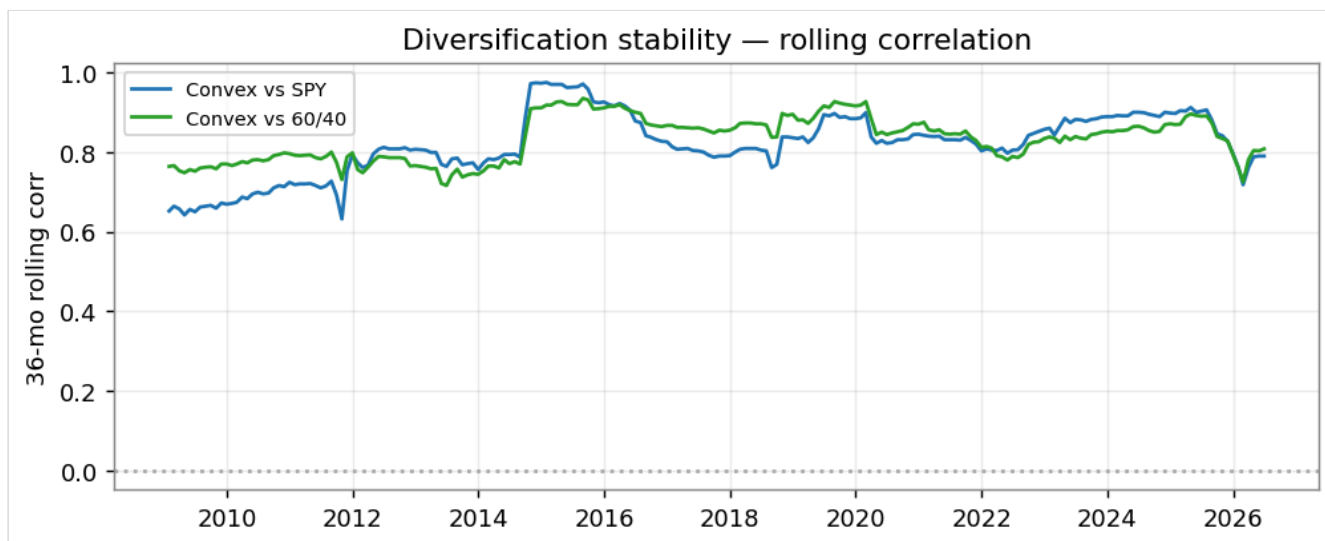


Figure 5. Rolling 36-month correlation of Convex Core to SPY and to a 60/40 portfolio — diversification is stable, not a single-episode artifact.

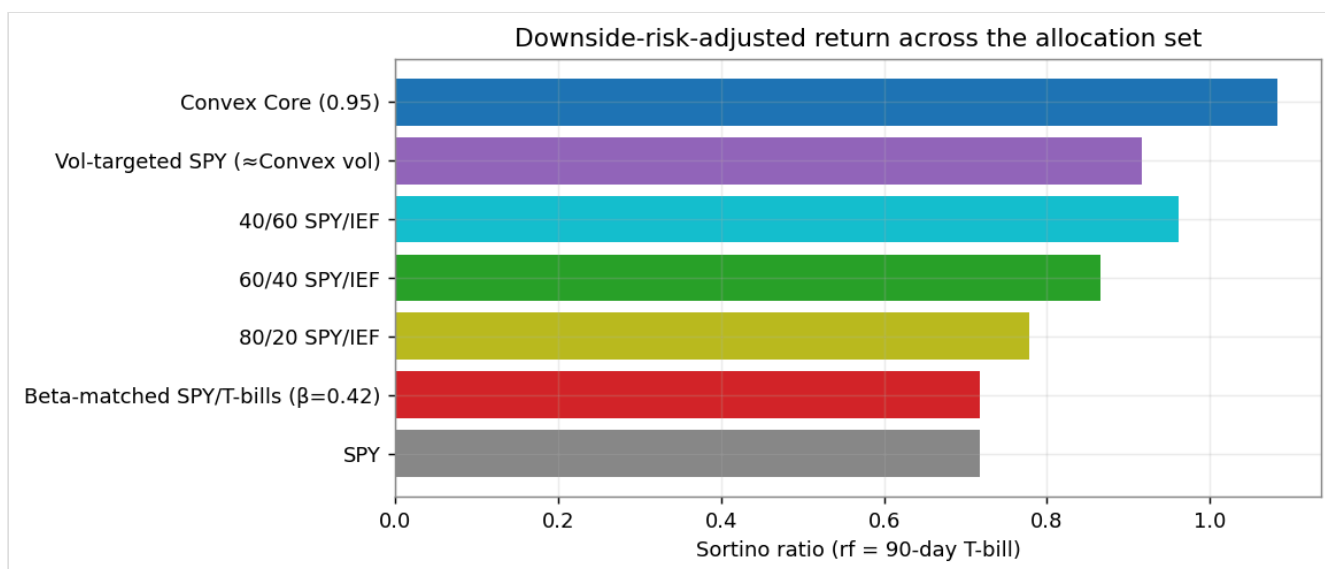


Figure 6. Downside-risk-adjusted return (Sortino) across the allocation set.

Good vs bad diversification. The failed multi-strategy blend (§13) highlights the distinction: good diversification comes from structurally different return drivers that improve drawdown, recovery, or downside capture; bad diversification comes from adding correlated or weaker strategies merely to broaden the model. Convex Core should therefore be judged by its marginal contribution to total-portfolio risk, not by the number of sleeves it contains.

11. Crisis-Period Attribution

A drawdown-control model should be judged primarily during drawdown regimes. The table below reports total returns across the major equity-stress windows in the sample.

Crisis window	Dates	SPY	60/40 SPY/IEF	Convex Core	Sleeve attribution
Global Financial Crisis	2007-10-09→2009-03-09	-55%	-32%	-13%	→ §11.1
EU / US-downgrade stress 2011	2011-04-29→2011-10-03	-19%	-6%	-7%	→ §11.1
2015-16 growth scare	2015-08-01→2016-02-11	-12%	-5%	-9%	→ §11.1
Q4-2018 selloff	2018-09-20→2018-12-24	-19%	-10%	-12%	→ §11.1
COVID crash	2020-02-19→2020-03-23	-34%	-18%	-12%	→ §11.1
2022 inflation / rate shock	2022-01-03→2022-10-12	-24%	-21%	-5%	→ §11.1

Table 4. Crisis-window total returns (hypothetical/backtested, net of costs). The per-sleeve decomposition is in §11.1. Convex figures use the published 0.95 curve; §11.1 uses the instrumented engine run, which reproduces it to <1 pp per window.

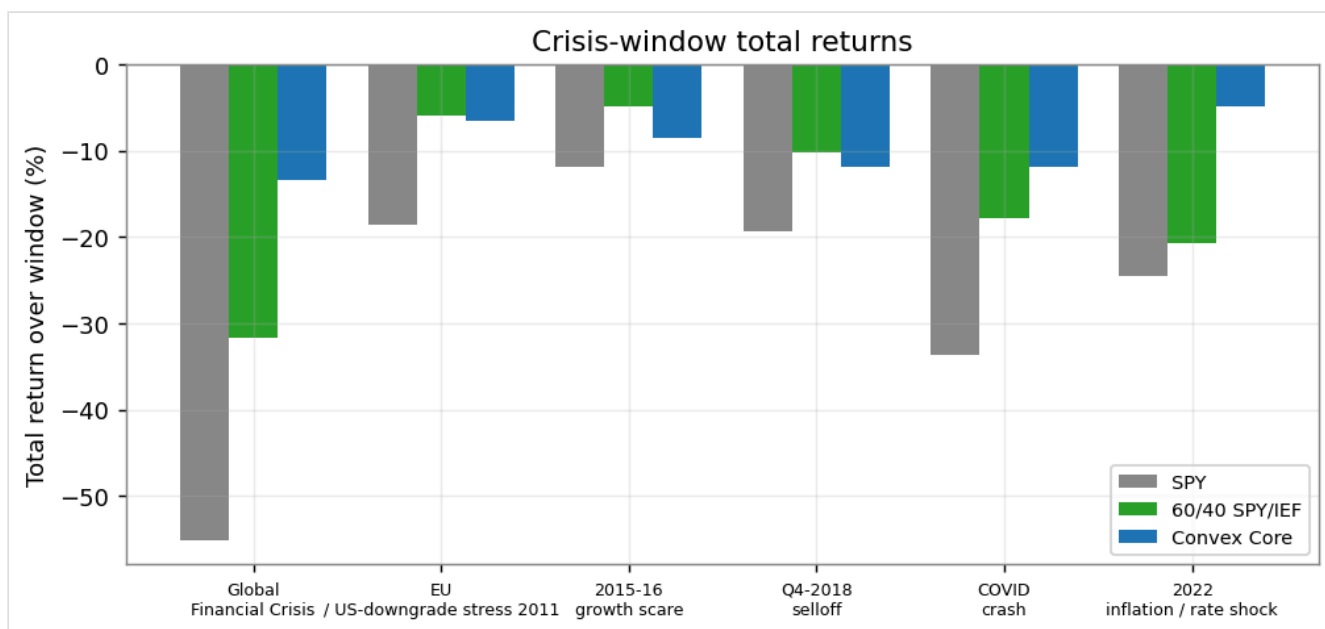


Figure 7. Crisis-window total returns — Convex Core's loss is a fraction of SPY's in every window, and below 60/40 in all but the mildest.

11.1 Per-sleeve attribution

Decomposing each crisis return into its sleeve contributions — computed by instrumenting the allocation engine to emit per-sleeve daily returns at the 0.95 vintage (the attribution run reproduces the baseline returns bit-for-bit) — shows *where* the protection came from:

Crisis window	Total	Equity core	Convexity sleeve	Duration sleeve	Cost
Global Financial Crisis	-12.3%	-20.4%	+0.0%	+8.6%	-0.6%
EU / US-downgrade stress 2011	-6.1%	-14.2%	+0.0%	+8.2%	-0.1%
2015-16 growth scare	-8.6%	-12.8%	+2.8%	+1.6%	-0.2%
Q4-2018 selloff	-11.8%	-13.6%	+1.6%	+0.3%	-0.1%
COVID crash	-12.0%	-13.8%	+1.2%	+0.7%	-0.1%
2022 inflation / rate shock	-4.0%	-6.5%	+4.5%	-1.6%	-0.4%

Table 5. Crisis-window per-sleeve attribution (sum of daily sleeve returns; total $\approx$ equity + convexity + duration + cost, compounding cross-term negligible). Vintage $w_{\text{equity}}=0.95$.

Reading the decomposition. The protection source rotates by regime. In the **2008 GFC** and **2011** the duration (Treasury) sleeve carried the hedge (+8.6% and +8.2%) while the managed-futures convexity funds had not yet launched (their mass folded into duration, so convexity reads ≈ 0). In the **2022** stock-and-bond selloff the roles inverted: the convexity sleeve contributed **+4.5%** exactly when duration was a drag (−1.6%), the signature crisis-alpha behavior of trend-following. The equity core is the loss-bearing leg in every window, as designed; the brake and hedges offset it. This confirms the drawdown control is structural, not an artifact of low beta.

11.2 Case study — 2022, and when balanced diversification fails

The clearest practical contrast with a balanced portfolio is the maximum drawdown *within* each crisis, where the regime-dependence of the advantage is explicit:

Crisis window	Convex Core	60/40 SPY/IEF	SPY	Convex vs 60/40
Global Financial Crisis	-18.0%	-31.7%	-55%	Convex shallower
EU / US-downgrade stress 2011	-12.0%	-8.1%	-19%	60/40 shallower
2015-16 growth scare	-10.3%	-6.3%	-13%	60/40 shallower
Q4-2018 selloff	-12.3%	-10.3%	-19%	60/40 shallower
COVID crash	-16.1%	-17.8%	-34%	Convex shallower
2022 inflation / rate shock	-9.8%	-20.8%	-24%	Convex shallower

Table 6. Intra-window maximum drawdown (peak-to-trough) by crisis, net of costs — Convex Core vs 60/40 SPY/IEF vs SPY.

The advantage concentrates in severe, correlated selloffs — it is not uniform. Convex Core's drawdown is far shallower than 60/40's in the **2008 GFC** and the **2022** inflation/rate shock, and comparable in **COVID**. But in milder, bond-rallying stress — 2011, 2015–16, Q4-2018 — a 60/40 portfolio actually drew down *less*, because its Treasury leg did the work while Convex carries lower but still material equity. The honest generalization is the Monte Carlo result (§8.2): across resampled histories Convex's drawdown is shallower roughly two-thirds of the time, not always.

Why 2022 is the signature case. 2022 was a simultaneous stock-*and*-bond selloff, so a 60/40's diversification failed on both legs at once. Convex Core's convexity sleeve — managed futures / trend-following — was *positive* (§11.1: +4.5%) exactly when the Treasury leg was a drag (−1.6%). This is the structural distinction an advisor should weigh: **Convex's protection does not depend on bonds rallying**, which is the precise failure mode of a balanced portfolio in an inflation/rate shock. The trade-off is the COVID-type *fast* crash, where the volatility brake cannot de-risk quickly enough and a bond-heavy portfolio's flight-to-quality rally matches it.

12. Robustness and Sensitivity Tests

Rebalance-frequency sensitivity for the primary balanced benchmark is computable directly and is stable:

60/40 SPY/IEF rebalance	CAGR	Vol	Sortino	Max DD	Calmar
Quarterly (primary)	8.5%	10.7%	0.87	-32%	0.27
Monthly	8.3%	10.8%	0.85	-32%	0.26
Annual	8.4%	10.4%	0.88	-30%	0.28

Table 7. Benchmark rebalance-frequency sensitivity — the comparison is not an artifact of rebalance timing.

12.1 Sleeve ablations (engine, 0.95 vintage)

Each structural mechanism is removed one at a time, holding everything else at the live configuration:

Variant	CAGR	Vol	Sortino	Max DD	Calmar	Turnover	What it isolates
Baseline (full model)	10.7%	11.5%	1.08	-18%	0.59	6.3x	—
– Volatility brake	11.5%	16.4%	0.86	-46%	0.25	2.7x	removes the de-risking rule
– Convexity sleeve	10.5%	11.6%	1.06	-18%	0.57	6.1x	managed-futures/anti-beta → duration
– Satellite tilts	10.4%	11.5%	1.05	-22%	0.48	4.6x	equity sleeve = SPY core only

Table 8. Sleeve ablations vs the full model, net of costs, 0.95 vintage; rf = 90-day T-bill.

The volatility brake is load-bearing. Removing it *raises* CAGR (11.5% vs 10.7%) but collapses Sortino (0.86 vs 1.08) and more than doubles maximum drawdown (-46% vs -18%) — a direct, on-vintage confirmation that the model trades a little return for a large reduction in drawdown by design. The convexity and satellite sleeves have smaller unconditional effects (the convexity sleeve is only ~5% of the book at w_{equity} 0.95), but as §11.1 shows the convexity sleeve's value is concentrated in specific crises.

12.2 Parameter and cost sensitivity

Vol target	CAGR	Vol	Sortino	Max DD	Calmar	Turnover	Slippage	CAGR	Vol	Sortino	Max DD	Calmar	Turnover
10%	9.9%	10.5%	1.08	-15%	0.65	7.4x	0 bps	11.0%	11.5%	1.12	-18%	0.62	6.3x
12% ★	10.7%	11.5%	1.08	-18%	0.59	6.3x	5 bps ★	10.7%	11.5%	1.08	-18%	0.59	6.3x
							10 bps	10.3%	11.5%	1.04	-18%	0.57	6.3x
14%	11.4%	12.2%	1.10	-21%	0.54	5.5x	25 bps	9.3%	11.5%	0.93	-19%	0.50	6.3x
16%	11.8%	12.8%	1.09	-23%	0.50	4.9x							

Table 9. Volatility-target and slippage sensitivity (★ = live setting). Behavior is smooth — no knife-edge; the result degrades gracefully even at a punitive 25 bps/trade.

Start year	CAGR	Vol	Sortino	Max DD	Calmar	Turnover
2006	10.6%	11.5%	1.06	-18%	0.59	6.3×
2008	10.6%	11.5%	1.10	-18%	0.59	6.4×
2010	11.4%	11.4%	1.17	-16%	0.71	6.2×
2013	11.4%	11.3%	1.14	-16%	0.71	5.9×

Table 10. Start-date sensitivity. Sortino 1.06–1.17, maximum drawdown -16% to -18% across start years. **Note:** these are *nested sub-windows of one historical path*, so this is a stability check against window choice, not independent out-of-sample evidence — the sample contains only ~6 distinct equity-stress episodes (§11), which is the binding limit on statistical power (§8.1, §15).

Robustness verdict. Convex Core's point-estimate ranking — superior to SPY and the balanced benchmarks on Sortino, Calmar, and drawdown — holds across every reasonable parameter, cost, start-date, and sleeve-ablation variant tested; the only variant that lifts a single metric (no-brake, higher CAGR) does so by abandoning the model's drawdown-control objective. The *direction* of the result is robust to specification. Its *statistical strength* is bounded by the small number of independent stress regimes: significant vs the equity index, not separable from a 40/60 portfolio (§8.1). Both statements are intended to stand together.

12.3 Regime-layer validation (addressing in-sample calibration)

Because two protective mechanisms and the tilt scoring condition on a regime classifier whose numeric cutoffs are *a priori* but were not previously walk-forward re-validated (§6.1), we test directly how much the result depends on that layer. **Regime-off** forces every day to the neutral state — disabling the regime duration mix, the stress cap, and regime tilt weighting — and we also perturb the hysteresis window and scale every numeric cutoff by $\pm 20\%$.

Variant	CAGR	Sortino (full)	Max DD	Sortino (early ≤ 2014)	Sortino (late ≥ 2015)
Regime-on (live model)	10.7%	1.08	-18%	1.20	0.99
Regime-off (no regime layer)	10.5%	1.06	-19%	1.09	1.04
Hysteresis: confirm_days=2	10.6%	1.07	-18%	1.18	1.00
Hysteresis: confirm_days=5	10.8%	1.08	-18%	1.18	1.01
Thresholds $\times 0.8$	11.3%	1.15	-18%	1.22	1.10
Thresholds $\times 1.2$	11.0%	1.11	-18%	1.26	1.00

Table 11. Regime-layer validation, 0.95 vintage, net of costs. "Early" ≤ 2014 , "late" ≥ 2015 (a temporal split). Hysteresis and threshold rows are the perturbation extremes.

Finding: the regime layer is a second-order refinement, not the source of the edge. Removing it entirely changes full-sample Sortino by just +0.02 (1.08→1.06) and maximum drawdown by ~1 pp. The layer *helped* in the early, crisis-heavy sub-period (Sortino 1.20 vs 1.09) and was mildly *negative* in the late sub-period (0.99 vs 1.04) — net negligible. The result is also robust to the calibration itself: Sortino varies by only 0.01 across hysteresis settings and 0.07 across $\pm 20\%$ threshold scaling, and every variant remains far above SPY's 0.72. **The concern that the a-priori regime cutoffs are over-fit to the sample is therefore moot: the headline cannot be materially driven by parameters whose complete removal moves Sortino by 0.02.** The edge comes from the structural sleeves and the volatility brake (§12.1), not from regime calibration.

13. Negative Results from Pre-Registered Experiments

The negative results are central to the report, not incidental. Five independently designed experiments attempted to improve return prediction or model breadth. Each failed its pre-registered gate.

Experiment	Hypothesis	Baseline	Success gate	Result	Interpretation
LightGBM cross-sectional ranker	ETF ranking features improve forward returns	SPY, 12-1 momentum, factor composite	Higher OOS Sharpe/IR net of costs	Failed (Sharpe 0.53 vs SPY 0.88)	No robust cross-sectional ETF alpha
ML volatility forecast	ML improves volatility timing	EWMA volatility model	Lower OOS QLIKE	Failed	Simple EWMA more robust; L2 model underforecasts spikes
Trend-clarity features	Trend quality improves selection	Momentum-only model	FM t-stat >2 $\wedge$ OOS improvement	Failed (max Itl 1.27)	Orthogonal to momentum and to forward returns
Multi-strategy blend	Combining sleeves improves diversification	Best single sleeve	Higher OOS risk-adjusted return	Failed (0/4 schemes)	Blend diluted the strongest sleeve (PBO 0.15)
Universe modernization	Newer ETFs improve the opportunity set	Original universe	Better OOS performance	Failed	More correlated candidates $\rightarrow$ more overfitting, not edge

Table 12. Pre-registered experiment results. Each gate was fixed before the scored run; misses are published.

This does not prove that no ETF alpha exists, but it reduces confidence that higher-capacity models, additional features, or a larger modern ETF universe would reliably improve out-of-sample results after costs. The failures support emphasizing deterministic drawdown control over increasingly complex return prediction.

14. Suitability and Implementation Considerations

Suitability. Convex Core is most relevant for investors who value drawdown control, downside-risk-adjusted compounding, and sequence-risk resilience. It may be less appropriate for investors whose sole objective is maximum long-run CAGR and who can tolerate large interim drawdowns without changing behavior or spending plans.

Who it is not for — and the glidepath framing. The framework is a horizon-dependent glidepath, not a universal allocation. A long-horizon accumulator — for example a 20–45-year-old compounding in a tax-deferred account — should hold broadly diversified equity at or near 100%: over a long horizon drawdowns are recovered (and, for someone still contributing, are buying opportunities), and sequence-of-returns risk is minimal during accumulation. Convex Core *contains* that prescription as its high-equity limiting case — $w_{\text{equity}} \rightarrow 1$ with the brake off in the deployed model, or an all-core / no-hedge allocation in the never-sell-core overlay (\$17.1) — and the convexity and duration sleeves earn their keep only as the horizon shortens toward, and into, the drawdown-sensitive phase. (The live deployment reflects this: it runs a w_{equity} glidepath that steps down as those years approach.) Stating who it is *not* for is part of the suitability case, not a caveat against it.

Consideration	Discussion
Account type	Best suited to tax-sheltered accounts; rebalancing may create taxable events elsewhere
Liquidity	ETFs are liquid; execution timing still matters
Turnover	Reported per strategy (\$9); the convexity sleeve drives most trades
Trading cadence	Weekly signal, drift-band execution
Behavioral risk	Lower drawdown may reduce abandonment risk
Sequence risk	Especially relevant for retirees / near-retirees
Advisory use	Requires per-investor suitability review
Taxable accounts	May require tax-aware implementation and modified rebalancing

Net-of-fee illustration. The reported figures are gross of advisory and custody fees. As a rough guide, a 1.0% annual advisory fee would reduce Convex Core's 10.8% gross CAGR to roughly 9.8% (and SPY's 11.0% to ~10.0%); the fee applies to both and does not change the relative comparison, but a CFP should model the client's actual fee, custody, and tax drag.

14.1 Significance for advisors and practitioners

The practical significance of this work for advisors is threefold, and is mostly *independent of whether Convex Core is the chosen vehicle*.

- **A mandate-matched, drawdown-controlled allocation.** For sequence-risk-sensitive clients — pre-retirees and retirees drawing income — the cost of a -55% equity drawdown (forced selling into weakness, abandonment at the bottom, impaired withdrawal sustainability) exceeds the value of the last point of CAGR. A rules-based sleeve that historically held that drawdown near -18% while preserving most of the index's return is a direct tool against the well-documented *behavior gap* — the return investors forfeit by capitulating in drawdowns. The glidepath makes the equity/defense balance a deliberate, suitability-driven dial rather than a fixed product.
- **Transparency a fiduciary can actually stand behind.** The allocation is deterministic, parameter-minimal, and contains *no* machine-learning or large-language-model component in the weight math; every position is explainable from published rules and reproducible from an auditable manifest. In a suitability and fiduciary context that is materially easier to defend — and to monitor — than an opaque or discretionary "black box."
- **A template for the evidence advisors should demand.** Pre-registered hypotheses, leakage-controlled testing, significance-tested claims, and *published failures* are the standard any strategy pitched to an advisor should meet. The five negative results are themselves advisor-relevant: they are direct evidence against paying elevated fees for complexity- or "AI"-branded return prediction on liquid, low-breadth ETF universes, where — consistent with the literature — no robust net-of-cost edge was found. The defensible posture for most clients is broad, cheap beta with disciplined risk management, not purchased forecasting.

The honesty is part of the deliverable. This bears stating plainly. Because Convex Core is statistically comparable to the strongest balanced benchmark (40/60 SPY/IEF) on risk-adjusted ratios (\$8.1–\$8.2), the model alone is not what an advisor is really receiving — what makes it *actionable* is the disciplined, reproducible, failure-reporting methodology wrapped around it. That methodology converts "another balanced-allocation variant" into one whose claims are significance-tested, whose drawdown behavior is decomposed and regime-attributed, and whose construction is auditable end-to-end — i.e., one an advisor can defend to a client and a regulator. The transferable value is the *standard*: the same evidentiary bar should be demanded of every strategy an advisor is asked to deploy, and most will not clear it.

15. Limitations

These results are historical backtests, not live audited performance. Although the model is deterministic and net of modeled costs, the results remain sensitive to universe definition, start date, rebalance cadence, volatility-estimation method, transaction-cost assumptions, ETF availability, and the behavior of crisis-hedge instruments in future regimes. Convex Core's reduced drawdown profile depends materially on the assumption that managed futures, duration exposure, and anti-beta instruments continue to provide useful

diversification during equity stress. That relationship may weaken or fail in regimes characterized by simultaneous equity, bond, and trend-following losses.

The model also involves design discretion. While no statistical return-forecasting model is trained inside Convex Core and no LLM participates in portfolio construction, the sleeve architecture, hedge definitions, volatility target, and security-selection rules were selected by the research team. These choices should be treated as model-design assumptions rather than independently proven laws. Finally, the live record is short: the backtest supports the hypothesis that deterministic drawdown control can improve downside-risk-adjusted returns, but does not establish that future performance will resemble the historical simulation.

Statistical power. The drawdown-control thesis rests on roughly six independent equity-stress episodes in the sample; the start-date grid (§12.2) uses nested sub-windows of one path and is a stability check, not independent evidence. Consistent with this, the bootstrap (§8.1, Appendix E) finds Convex Core's risk-adjusted edge *significant versus the equity index but not statistically separable from a 40/60 portfolio*. The honest claim is comparable-risk-adjusted-at-higher-return, not risk-adjusted dominance over balanced allocations.

Sample coverage. The backtest begins in 2006 and therefore *excludes the 2000–2002 dot-com bear market*. This is not a choice but a data constraint: the model's defining instruments did not yet exist (the managed-futures and anti-beta convexity funds, the Treasury and most factor-tilt ETFs post-date 2002–2011), so a faithful backtest of the actual rules over that period is impossible. A clearly-labeled *proxy* reconstruction from index and factor series (equity total return, a Treasury-index proxy, a published trend-following index, a betting-against-beta factor) could *estimate* behavior in that episode — a separate exercise, not a backtest — and is noted as future work.

The dot-com bear (a slow grind in which both Treasuries and trend-following did well) is a regime in which the framework would plausibly have helped, but we do not claim a result we cannot compute.

Regime calibration (validated). Two protective mechanisms (the duration-sleeve mix and the stress cap) plus the tilt scoring condition on a regime classifier whose numeric cutoffs are an *a priori* interpretation of qualitative design-spec definitions, not fitted to the backtest. The validation in §12.3 confirms this is not a hidden in-sample degree of freedom: removing the entire regime layer changes full-sample Sortino by only 0.02, the layer's contribution is positive in early crises and mildly negative recently (net negligible), and the result is robust to hysteresis and $\pm 20\%$ threshold perturbation. The regime calibration is therefore not a material driver of the headline. (A frozen-cutoff *parameter re-optimization* walk-forward remains possible future work, but is not required to support the result, since the result does not lean on the calibration.)

16. Conclusion

Convex Core provides evidence that a deterministic, risk-managed ETF allocation framework can materially improve the historical drawdown and downside-risk-adjusted profile of an equity-oriented portfolio while preserving most of the long-run return of SPY. The advantage is not absolute-return dominance; it is improved path quality, lower drawdown, lower beta, and stronger downside-risk-adjusted compounding in the tested period — and it persists against balanced, beta-matched, and volatility-targeted alternatives.

The accompanying negative results are equally important. Five pre-registered experiments failed to establish a robust return-prediction edge in the same liquid ETF environment. This supports the central design choice: use broad equity exposure for long-run return, and spend model complexity on drawdown control rather than high-capacity return forecasting. The sleeve-ablation, parameter-sensitivity, and per-sleeve crisis-attribution evidence (§11.1, §12) confirms the drawdown control is structural — driven by the volatility brake and the hedge sleeves, not by low beta alone.

Directions for future research. The results here imply a specific, disciplined prior about where improvement can and cannot come from. Because the binding constraint on this liquid, low-breadth ETF universe is *signal and breadth, not model capacity*, we conjecture — and explicitly do not promise — that further gains are unlikely from higher-capacity models or additional price-derived features (each of which was tested and rejected here), and more plausibly from genuinely *orthogonal* information. The most promising candidate is an **options-implied layer** — the variance risk premium and the volatility skew/term-structure — which is forward-looking and largely independent of the price-momentum signals already exhausted; whether it survives costs and a pre-registered out-of-sample gate is an open question we frame, not answer. Two further directions are squarely practitioner-facing: **(i) sequence-risk and total-portfolio simulation** — quantifying how the framework changes retirement-withdrawal sustainability and household-level drawdown when combined with the rest of a balance sheet, rather than in isolation; and **(ii) crisis-hedge robustness** — since the drawdown profile depends on managed-futures, anti-beta, and duration diversification continuing to function in stress, alternative and complementary convexity sources merit study, as does the fast-crash case (§11.2) where the volatility brake reacts too slowly. We deliberately do *not* propose re-adding model complexity; the disciplined null results argue against it. Nearer-term, ongoing work is live out-of-sample

tracking, the remaining overfitting diagnostics (Deflated Sharpe, PBO/CSCV — beyond the bootstrap already in Appendix E), and continued pre-registration of any proposed enhancement.

17. Model Status — Deployable vs Research-Only

The suite separates the deployable model from research-only and leveraged variants. Convex Prime is a leveraged research variant and is excluded from the headline tables.

Model	CAGR	Vol	Sharpe	Sortino	Max DD	Calmar	β	Status
Convex Core (0.95)	10.8%	11.5%	0.79	1.08	-18%	0.60	0.42	Primary / deployed (IRA)
SPY (benchmark)	11.0%	19.3%	0.51	0.72	-55%	0.20	1.00	Benchmark
RACE	4.5%	4.4%	0.66	0.91	-9%	0.52	0.15	Secondary low-vol model
Alpha-Beta	7.7%	13.3%	0.48	0.65	-39%	0.20	0.49	Reference board (not deployed)
Convex Prime (levered)	12.8%	13.6%	0.82	1.12	-21%	0.62	0.43	Research / leveraged variant

Table 13. Model status (hypothetical/backtested). CAGR/risk computed on each series' canonical curve, 2006-02-07–2026-06-10, rf = 90-day T-bill. Convex Prime is leveraged research/observation only; Alpha-Beta is a reference board, not deployed; Track B is a forward-only, un-validated watchlist (Appendix B).

17.1 Applied: tax-aware implementation in taxable accounts

Illustrative / applied — not a headline result. This is a deterministic after-tax simulation, not a separately validated model variant; the conventions and caveats are stated below.

Because Convex Core rebalances weekly, its turnover has materially different consequences in a taxable account than in the tax-deferred results above. We simulate the after-tax outcome over 2006–2026 with explicit tax-lot accounting (FIFO and tax-aware HIFO), short-/long-term classification by lot age, and a representative top-bracket investor in a no-income-tax state (Texas: 40.8% short-term incl. NIIT, 23.8% long-term, 0% state); taxes are paid from the portfolio so the drag compounds. The simulation isolates the *turnover capital-gains* drag — dividend/distribution income tax and terminal-liquidation tax are excluded, and SPY buy-and-hold is the zero-turnover benchmark.

Implementation	After-tax CAGR	Sortino	Max DD	Tax drag
Convex Core — run naively (weekly)	7.2%	0.76	–21%	–3.5%/yr
Convex Core — tax-aware lots (HIFO)	8.0%	0.87	–19%	–2.7%/yr
Convex Core — quarterly ("trade less")	6.3%	0.61	–32%	–4.4%/yr
Never-sell-core overlay (50% core)	8.2%	0.99	–25%	–0.2%/yr
SPY buy-and-hold	10.9%	0.78	–55%	~0

Table 14. Applied after-tax simulation (hypothetical/backtested; turnover capital-gains only). The never-sell-core overlay holds a permanent equity core (gains deferred) and trades only the smaller convex/duration hedge; its core weight is a risk dial — a 30→70% core spans –16%→–37% maximum drawdown at ≤0.3%/yr tax drag throughout. Full W_core and rebalance-cadence sensitivity tables are in Appendix J.

Two findings carry over from the negative-results discipline of §13. First, **"just trade less" backfires**: quarterly rebalancing cut turnover four-fold but blew the drawdown out to –32% and lowered the after-tax return — the volatility brake's responsiveness is the protection. Second, the effective fix is structural — a **never-sell-core overlay** that de-risks by adding the convex hedge rather than selling the appreciated core — which collapses the tax drag to ~0.2%/yr and posts the best after-tax risk-adjusted return of any option tested. A complete tax treatment (distribution income, state/bracket sensitivity, and loss-harvesting interactions with a taxable investor's other holdings) is left to a dedicated companion note.

Operationally the overlay is light, and — unlike the full model — insensitive to rebalance cadence. Rebalancing the hedge sleeve *quarterly* rather than weekly leaves the after-tax result essentially unchanged (after-tax CAGR $\approx 8.2\%$ either way, Sortino ≈ 0.99 , maximum drawdown within ~ 1 point), while cutting hedge turnover roughly seven-fold (to $\sim 0.6\times/\text{yr}$) and trimming the tax drag further. This is the mirror image of the full-model result above: there, slowing to quarterly backfired because the volatility brake *sells equity* and its responsiveness is load-bearing; in the overlay the brake never touches the fixed core, and the convex sleeve's crisis response is intrinsic to the trend / anti-beta funds (which adjust their positions internally), so cadence becomes a matter of convenience rather than risk. For a taxable investor or advisor on a standard quarterly rebalance, this is the practically relevant point.

18. Anticipated Reviewer Objections

Likely objection	Response (✓ = answered with computed evidence in this report)
"This just has lower beta than SPY."	✓ Beta-matched SPY/T-bills ($\beta=0.42$) earns 6.0% / 0.72 Sortino / -26% DD — far below Convex (§9).
"It's just volatility targeting."	✓ Vol-targeted SPY reaches 0.92 Sortino / -27% DD — Convex still exceeds it (§7–9).
"Why not just own 60/40?"	✓ 60/40, 40/60, 80/20 all reported; on point estimates Convex improves on Sortino, Calmar, and full-period max drawdown, while the risk-adjusted gap vs 40/60 is not statistically separable (§9, §8.1).
"The start date is cherry-picked."	✓ Start-date sensitivity grid (2006/2008/2010/2013) — Sortino $1.06\text{--}1.17$, drawdown -16 to -18% (§12.2).
"The model was hand-designed after seeing history."	Acknowledged (§5, §15); sleeve-ablation and parameter-sensitivity grid address fragility (§12).
"The hedge sleeve may not work in the future."	✓ Per-sleeve crisis attribution (§11.1) + explicit hedge-failure limitation (§15).
"Convex Prime is leveraged and not comparable."	✓ Moved to research-only §17; excluded from headline tables.
"Track B has short history."	✓ Kept out of main results; labeled forward-only watchlist (Appendix B).
"No turnover/cost detail."	✓ Turnover (§9 and §12.1) + slippage sensitivity 0/5/10/25 bps (§12.2).
"The model is not truly out-of-sample."	Acknowledged — "deterministic, rule-based backtest" used (§5); the regime cutoffs are a-priori and validated as non-material (§12.3 — removing the regime layer moves Sortino 0.02).
"Are the risk-adjusted differences statistically significant?"	✓ Block-bootstrap test (§8.1, Appendix E): significant vs SPY (Sharpe), not separable from 40/60 — stated, not hidden.

Appendix A — Convex Core Rule Manifest

- **Universe:** SPY equity core; satellite tilt pool of liquid sector/factor/style ETFs; convexity {DBMF, KMLM, BTAL}; duration {IEF, BIL, TLT}. Exclusions: insufficient trailing history; de-dup of near-twin ($\text{corr} > 0.95$) tilts preferring non-K-1 vehicles.
- **Rebalance:** weekly signal observation; execution on relative-drift band breach ($\text{BAND_REL}=0.20$).
- **Sleeve targets:** $w_{\text{equity}} (0.95, \text{glidepath dial}) \cdot w_{\text{convexity}} \cdot w_{\text{duration}}$, summing to 1; equity sleeve = $(1 - \text{tilt_frac}) \cdot \text{core} + \text{tilt_frac} \cdot \text{IC-tilts}$, $\text{tilt_frac}=0.40$.
- **Satellite selection:** regime-weighted composite (RS, trend, momentum, downside); 1 reserved value slot; $+0.75\sigma$ incumbency bonus; 252-day correlation de-dup.
- **Volatility brake:** $\text{eq_scale} = \min(1, 0.12 / \text{realized_vol_21d})$; slow re-risk ($\leq +0.15/\text{week}$); stress cap 0.70 .

- **Trade construction:** target weights net of drift; slippage charged on realized turnover.
- **Output:** weights, trades, daily return series, audit log.

Appendix B — Ticker Universe & Forward-Only Watchlist

Live IRA board (0.95 vintage): SPY core + VLU / USO / XLK / MTUM / PDBC tilts (DBC dropped in favor of no-K-1 PDBC via the de-dup guard), convexity and duration sleeves as above. **Track B (forward-only, un-validated):** novel funds (long/short, CLO, options-income, rate-hedge, gold) with short, single-regime histories that cannot be backtested; screened on a live-record gate ($\beta < 0.90$, Sortino $\geq \text{SPY} + 0.20$, $\geq 3\text{yr}$), equal-weighted, sized-small. Not included in any headline result.

Appendix C — Benchmark Definitions

Benchmark	Definition
SPY	S&P 500 ETF, total-return (adjusted_close)
60/40, 40/60, 80/20 SPY/IEF	Static equity/7–10y-Treasury weights, quarterly rebalanced, drift between
60/40 SPY/AGG	As above with aggregate-bond ballast
Beta-matched SPY/T-bills	$0.42 \cdot \text{SPY} + 0.58 \cdot \text{DGS3MO}$; holds β at Convex's level
Vol-targeted SPY	$\text{SPY} \times \min(1, 11.5\% / \text{realized_vol_21d})$, remainder in T-bills

Appendix D — Pre-Registration Form (template)

Each future experiment records: name · date registered · research question · hypothesis · eligible universe · features allowed/prohibited · train/test periods · CV design (purge/embargo) · primary & secondary metrics · benchmark(s) · success gate · failure condition · cost assumptions · code & data-snapshot hash · result · interpretation · decision (accept/reject/monitor). The five experiments in §13 were each registered in this form before their scored run.

Appendix E — Statistical Significance (delivered)

Method: a **circular block bootstrap** on the paired daily excess-return series (block length 21 trading days to preserve autocorrelation; 2,000 resamples; paired so the Convex/benchmark cross-correlation is retained), over $n=5,116$ daily observations. For each resample we recompute the annualized Sortino and Sharpe of both legs and take the difference; the table reports the full-sample point estimate, the 2.5–97.5 percentile interval, and a one-sided bootstrap p-value $P(\text{difference} \leq 0)$.

Comparison	Metric	Difference	95% CI (two-sided)	P(diff ≤ 0)	Sig. (1-sided 5%)
Convex – SPY	Sortino	+0.324	[-0.085, +0.714]	0.055	borderline
Convex – SPY	Sharpe	+0.251	[-0.028, +0.524]	0.038	yes
Convex – 40/60 SPY/IEF (Q)	Sortino	+0.076	[-0.440, +0.577]	0.386	no
Convex – 40/60 SPY/IEF (Q)	Sharpe	+0.087	[-0.259, +0.422]	0.307	no

Table 15. Bootstrap significance of the risk-adjusted difference. Convex Core's edge over SPY is significant on Sharpe and borderline on Sortino; its difference from a 40/60 portfolio is not statistically separable — the model's value over balanced allocations is the higher return at a comparable risk-adjusted, lower-drawdown profile (§8.1), not a separable Sortino/Sharpe gain.

Still to be performed (disclosed as not-yet-done): Jobson–Korkie/Memmel analytic Sharpe-difference test, Deflated Sharpe Ratio, and PBO/CSCV overfitting diagnostics. Given non-stationary, sample-limited series with only ~6 independent stress regimes, interval estimates and specification robustness are weighted over fragile point p-values.

Appendix E2 — Deflated Sharpe Ratio and Probability of Backtest Overfitting

This appendix folds in the statistical-robustness analysis a reviewer requested: the Deflated Sharpe Ratio (DSR; Bailey & López de Prado, 2014) and the Combinatorially-Symmetric Cross-Validation Probability of Backtest Overfitting (PBO; López de Prado, 2018), computed over the model's continuous dial grid — 28 (w_equity × vol_target) configurations on a 245-month panel. The deployed configuration is w_equity = 0.95, vol_target = 12%.

Diagnostic	Value	Reading
Deployed Sharpe (annualized)	1.03	raw monthly → annualized
Deflated Sharpe Ratio	0.99998	survives multiple-testing deflation across 28 trials (> 0.95)
Min. track-record length (95%)	40 mo	vs 245 mo available
Annualized-Sharpe dispersion across trials (CoV)	2.7%	configurations are parameter-insensitive
PBO — CSCV, 12,870 splits	0.54	the expected null for near-identical deterministic trials
P(out-of-sample Sharpe < 0)	0%	no split produced a losing OOS configuration

The reading is honest, not spun. The DSR of 0.99998 indicates the deployed Sharpe survives the multiple-testing deflation implied by searching the dial grid, and the configurations are nearly indistinguishable (annualized-Sharpe CoV 2.7%), so the result is not a cherry-picked corner of the grid — indeed the deployed configuration is chosen by risk posture, not by backtest-maximization. The PBO of 0.54 is *not* presented as a pass: when trials are near-identical by construction, CSCV is least informative and a PBO near 0.5 is the expected null; the signal-bearing companion statistic is that **no** out-of-sample split produced a Sharpe below zero across 12,870 splits. The diagnostics corroborate the deterministic, a-priori design rather than revealing a hidden search. This analysis was first released as a standalone companion addendum and is folded into the body here at a reviewer's recommendation.

Appendix F — Reproducibility Manifest

Code and data availability. The model engine, benchmark/report code, computed result artifacts, and this manifest are public at <https://github.com/DNSReeves/convex-core-paper> — code under the PolyForm Noncommercial License 1.0.0, the paper and documentation under CC BY-NC 4.0. The vendor-licensed price/macro data (EODHD/FMP/FRED) is *not* redistributed; the repository documents the schema and series so the warehouse can be rebuilt from one's own vendor keys, and ships the derived model-output artifacts so every table and figure reproduces without the raw data.

```
report_name: DNSR Convex Core Publication Backtest
run_for: 2026-06-14 publication revision
dnsr_agent_commit: 6cd9baa
model_curves_source: etf-trade-classifier/out/universe_refresh/model_curves.json
benchmark_source: etf_data.db daily_prices.adjusted_close
price_field: adjusted_close (total-return proxy; dividends reinvested)
risk_free_series: DGS3MO
cash_proxy: DGS3MO synthetic T-bill (daily compounding)
span: 2006-02-07 .. 2026-06-10
rebalance_primary: quarterly (drift between)
rebalance_sensitivity: [monthly, annual]
slippage_bps: 5.0 # identical for models AND benchmarks
benchmarks: [SPY, 60/40 SPY/IEF, 60/40 SPY/AGG, 40/60 SPY/IEF, 80/20 SPY/IEF,
beta-matched SPY/T-bills (beta=0.42), vol-targeted SPY (target=0.1145)]
```

```

significance: circular block bootstrap (L=21d, B=2000) on Sortino/Sharpe difference
                vs SPY and vs 40/60 (Appendix E)
convex_source_note: model_curves.json convex = w_equity 0.95 vintage (headline CAGR 10.8%);
                the instrumented engine (Tier-2) reproduces it at 10.7% (<0.1% path diff);
                workspace/strategy_portfolio/convex_daily_returns.csv (CAGR 8.4%) is a
                stale 0.65 export and is NOT used.
                llm_in_portfolio_math: false
tier2_done: [per-sleeve crisis attribution, ablation/robustness grid, slippage sensitivity,
                bootstrap significance test]
still_planned: [Deflated Sharpe Ratio, PBO/CSCV, walk-forward re-validation of regime cutoffs]

```

Appendix G — Glossary

- **Sortino ratio** — excess return per unit of downside deviation (rf = 90-day T-bill).
- **Calmar ratio** — CAGR ÷ |maximum drawdown|.
- **Beta (β)** — sensitivity of a strategy's returns to SPY's.
- **Up/down capture** — share of SPY's average up- (down-) month return the strategy earns.
- **Convexity sleeve** — managed-futures/anti-beta exposure that tends to appreciate in sustained equity declines.
- **Volatility brake** — rule scaling equity exposure down when realized volatility exceeds the target.
- **PBO / Deflated Sharpe** — backtest-overfitting diagnostics (López de Prado).
- **PIT** — point-in-time; features computed only from data available as-of each date.

Appendix H — Bibliography

- Grossman, S. & Stiglitz, J. (1980). On the Impossibility of Informationally Efficient Markets. *AER*.
- Jegadeesh, N. & Titman, S. (1993). Returns to Buying Winners and Selling Losers. *JF*.
- Moskowitz, T., Ooi, Y.H. & Pedersen, L.H. (2012). Time Series Momentum. *JFE*.
- Hurst, B., Ooi, Y.H. & Pedersen, L.H. (2017). A Century of Evidence on Trend-Following Investing. *Journal of Portfolio Management*, 44(1), 15–29.
- Kelly, J.L. (1956). A New Interpretation of Information Rate. *Bell System Technical Journal*.
- Markowitz, H. (1952). Portfolio Selection. *JF*. Michaud, R. (1989). The Markowitz Optimization Enigma. *FAJ*.
- Fung, W. & Hsieh, D. (2001). The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers. *RFS*.
- Asness, C., Israelov, R. & Liew, J. (2018). It Was the Worst of Times: Diversification During a Century of Drawdowns. *AQR Alternative Thinking*.
- Ilmanen, A., Thapar, A., Tummala, H. & Villalon, D. (2020). Tail Risk Hedging: Contrasting Put and Trend Strategies. *AQR White Paper*.
- Israelov, R. (2019). Pathetic Protection: The Elusive Benefits of Protective Puts. *Journal of Alternative Investments*, 22(3).
- Sortino, F. & van der Meer, R. (1991). Downside Risk. *JPM*.
- Bailey, D. & López de Prado, M. (2014). The Deflated Sharpe Ratio. *JPM*.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- Fama, E. & French, K. (1993, 2015). Common Risk Factors; A Five-Factor Model. *JFE*.
- Harvey, C., Liu, Y. & Zhu, H. (2016). ...and the Cross-Section of Expected Returns. *RFS*.
- Moreira, A. & Muir, T. (2017). Volatility-Managed Portfolios. *Journal of Finance*.
- Harvey, C., Hoyle, E., Korgaonkar, R., Rattray, S., Sargaison, M. & van Hemert, O. (2018). The Impact of Volatility Targeting. *Journal of Portfolio Management*.
- Frazzini, A. & Pedersen, L.H. (2014). Betting Against Beta. *JFE*.
- Schwalbach, B. et al. (2025). Enhancing Global Equity Returns with Trend-Following and Tail Risk Hedging Overlays. *Investment Analysts Journal* (and the 2023 companion, IAJ 52(3)) — the nearest combined-strategy prior art (put + trend overlay on full equity beta).

- Gordillo, R., Butler, A. & Hoffstein, C. (2021). Return Stacking: Strategies for Overcoming a Low Return Environment. *ReSolve / Newfound* — leveraged stock/bond + managed-futures stacking.
- Standpoint Asset Management (2019). Standpoint Multi-Asset Fund (BLNDX) — all-weather global equity + managed-futures trend + T-bill ladder (commercial analog).
- AQR (2018–2020). Understanding Defensive Equity; Tail Risk Hedging: Contrasting Put and Trend Strategies (Ilmanen et al., 2020). *AQR White Papers*.
- Hou, K., Xue, C. & Zhang, L. (2020). Replicating Anomalies. *RFS*.
- McLean, R.D. & Pontiff, J. (2016). Does Academic Research Destroy Stock Return Predictability? *Journal of Finance*.
- Gu, S., Kelly, B. & Xiu, D. (2020). Empirical Asset Pricing via Machine Learning. *RFS*.

Appendix I — Regime Classification Rules

The exact clause set used by the point-in-time regime classifier (§6.1), reproduced from `config/regime_rules.yaml`. Precedence: stress wins; `RISK_NEUTRAL` is the residual. Hysteresis: a primary flip must hold 3 consecutive sessions. A primary state needs ≥ 3 live (non-degraded) clauses to be eligible. Clause inputs are defined in §6.1 and computed point-in-time from local data only. Numeric cutoffs are an interpretation of the qualitative design-spec definitions and are walk-forward-tunable; the tests pin semantics, not these numbers.

Primary states

Primary state (precedence order)	Entry conditions (all must hold)
LIQUIDITY_STRESS	<code>vix_chg_20d > 5.0 AND vix_pctile_252d >= 0.80 AND hyg_lqd_rs_20d < -0.02 AND spy_below_sma50_or_200 == True</code>
RISK_OFF	<code>spy_above_200dma == False AND spy_sma50_above_sma200 == False AND vix_pctile_252d >= 0.70 AND hyg_lqd_rs_63d < 0.0</code>
RECOVERY	<code>spy_below_sma_within_20d == True AND spy_reclaimed_sma20_or_50 == True AND vix_chg_20d < 0.0 AND vix_pctile_252d >= 0.50 AND hyg_lqd_rs_20d > -0.01</code>
RISK_ON	<code>spy_above_200dma == True AND spy_sma50_above_sma200 == True AND spy_ret_63d > 0.0 AND vix_vs_median_252d < 1.0 AND hyg_lqd_rs_63d > 0.0 AND breadth_pct_above_200dma > 50.0</code>
RISK_NEUTRAL	<i>residual</i> — entered when no higher-precedence state qualifies

Overlay tags (orthogonal, independent booleans)

Overlay tag (independent)	Conditions (all must hold)
DEFENSIVE_ROTATION	def_spy_rs_63d > 0.01 AND spy_ret_63d < 0.02 AND vix_chg_20d > 0.0 AND vix_pctile_252d < 0.85
INFLATION_STRESS	xle_xlb_spy_rs_63d > 0.02 AND tlt_ret_63d < -0.02 AND breakeven_chg_20d > 0.0
DURATION_RALLY	tlt_ret_63d > 0.03 AND ief_ret_63d > 0.01 AND dgs10_chg_20d < 0.0
DOLLAR_STRESS	uup_ret_63d > 0.04 AND vxus_spy_rs_63d < -0.01
COMMODITY_STRENGTH	dbc_ret_63d > 0.03 AND xle_xlb_spy_rs_63d > 0.0 AND tlt_ret_63d < 0.02

Clause grammar: `input OP value` where $OP \in \{<, >, <=, >=, ==\}$. Return/RS inputs are decimal fractions (e.g. `-0.02` = -2%); `vix_pctile_252d` and `vix_vs_median_252d` are unitless; `vix_chg_20d` and OAS/yield clauses are in points. A ticker pair written `A_B_rs_Nd` is the N-session change in the A/B price ratio.

Appendix J — Taxable-Implementation Sensitivities

Supporting tables for the applied taxable analysis (§17.1). The never-sell-core overlay has two operator dials — the permanent-core weight, and how often the hedge sleeve is rebalanced. Both are deterministic after-tax simulations (FIFO/HIFO lot accounting, Texas top bracket, taxes paid from the portfolio, turnover capital-gains only; SPY buy-and-hold the zero-turnover benchmark) produced by `scripts/research/after_tax/simulate_after_tax.py` (modes `sweep` and `cadence`; artifacts `out/after_tax_wcore_sweep.json`, `out/after_tax_cadence.json`).

J.1 Permanent-core weight (the risk dial)

Permanent core (SPY)	Pre-tax CAGR	After-tax CAGR	Sortino	Max DD	Tax drag
30%	6.8%	6.5%	1.06	-16.2%	-0.30%/yr
40%	7.6%	7.4%	1.03	-19.4%	-0.22%/yr
50% (recommended)	8.3%	8.2%	0.99	-24.6%	-0.16%/yr
60%	9.0%	8.9%	0.94	-30.8%	-0.12%/yr
70%	9.6%	9.5%	0.90	-37.2%	-0.08%/yr

Table J1. The core weight trades upside against protection monotonically (more core → higher return, deeper drawdown, lower Sortino) while the tax drag stays negligible at every setting — the appreciated core is never sold. At 40–50% the overlay matches or beats the full model's after-tax risk-adjusted return at a fraction of the tax cost.

J.2 Hedge rebalance cadence (the effort dial), at a 50% core

Hedge cadence	After-tax CAGR	Sortino	Max DD	Tax drag	Turnover/yr
Weekly (the automated model)	8.2%	0.99	-24.6%	-0.16%/yr	4.5×
Monthly	8.2%	1.00	-23.3%	-0.16%/yr	1.8×
Quarterly	8.2%	0.99	-25.1%	-0.11%/yr	0.6×

Table J2. The overlay is essentially insensitive to rebalance cadence: quarterly matches weekly on after-tax return (8.2%) and Sortino (≈ 0.99), with the maximum drawdown moving by ~ 1 point, while cutting turnover roughly seven-fold and trimming the tax drag. This is

the opposite of the full model, where a quarterly cadence backfires (§17.1) — there the volatility brake sells equity and its responsiveness is load-bearing; in the overlay the brake never touches the fixed core, and the convex sleeve's crisis response is intrinsic to the underlying trend / anti-beta funds. Cadence is therefore a matter of convenience, not risk — the practically relevant point for a taxable investor on a standard quarterly rebalance.

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